



# **NRC7394 Evaluation Kit**

## **User Guide**

### **(J-Link Setup and Debugging)**

**Ultra-low power & Long-range Wi-Fi**

**Ver 1.0**  
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**NEWRACOM, Inc.**

## **NRC7394 Evaluation Kit User Guide (J-Link Setup and Debugging)** **Ultra-low power & Long-range Wi-Fi**

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# 1 Introduction

This document describes how to install, configure, and use the SEGGER J-Link debugger with NRC series chipsets in standalone mode.

It provides detailed instructions for:

- Installing required tools
- Setting up Ozone IDE
- Connecting hardware via SWD
- Downloading firmware to RAM and external flash
- Using script-based control for Serial Flash operations

## 2 Debugger Requirement

### 2.1 Hardware Requirement

- Debugger : SEGGER J-Link Plus or higher version
- Target Board: NRC series chipsets



Figure 2.1 J-Link Plus

### 2.2 SWD Connection

Based on the standard J-Link 20-pin SWD interface, connect the signals as follows:

|             |    |                          |                          |    |                |
|-------------|----|--------------------------|--------------------------|----|----------------|
| VCC (VTref) | 1  | <input type="checkbox"/> | <input type="checkbox"/> | 2  | VCC (optional) |
| N/U         | 3  | <input type="checkbox"/> | <input type="checkbox"/> | 4  | GND            |
| N/U         | 5  | <input type="checkbox"/> | <input type="checkbox"/> | 6  | GND            |
| SWDIO       | 7  | <input type="checkbox"/> | <input type="checkbox"/> | 8  | GND            |
| SWCLK       | 9  | <input type="checkbox"/> | <input type="checkbox"/> | 10 | GND            |
| N/U         | 11 | <input type="checkbox"/> | <input type="checkbox"/> | 12 | GND            |
| SWO         | 13 | <input type="checkbox"/> | <input type="checkbox"/> | 14 | GND            |
| RESET       | 15 | <input type="checkbox"/> | <input type="checkbox"/> | 16 | GND            |
| N/C         | 17 | <input type="checkbox"/> | <input type="checkbox"/> | 18 | GND            |
| N/C         | 19 | <input type="checkbox"/> | <input type="checkbox"/> | 20 | GND            |

|                                                  |                                                                     |
|--------------------------------------------------|---------------------------------------------------------------------|
| <span style="color: red;">■</span> 1 VTref (VCC) | <span style="color: blue;">■</span> 9 SWCLK                         |
| <span style="color: green;">■</span> 7 SWDIO     | <span style="color: black;">■</span> 10 GND (Connect to Target GND) |

| Pin No. | Signal      | Description                                 |
|---------|-------------|---------------------------------------------|
| 1       | VTref (VCC) | Target reference voltage (3.3V recommended) |
| 7       | SWDIO       | Serial Wire Debug I/O                       |
| 9       | SWCLK       | Serial Wire Debug Clock                     |
| 10      | GND         | Ground (Connect to Target GND)              |

Figure 2.2 J-Link SWD Connection Guide

- Pin 1 (VCC / VTref) → Target VCC
- Pin 7 (SWDIO) → Target SWDIO (GPIO 10 in NRC7394)
- Pin 9 (SWCLK) → Target SWCLK (GPIO 11 in NRC7394)
- Pin 10 (GND) → Target GND

## 2.3 Software Requirement

- NRC Materials
  - nrc7394\_boot\_config.zip  
Writes debug configuration (e.g., enable J-Link debug, disable WDT, skip CRC)
  - J-Link device support files (FlashNRCXXX.elf, JLinkDevices.xml)  
Enable device recognition and external flash programming
- IDE: SEGGER Ozone Debugger  
Used for firmware debugging (breakpoints, step execution, inspection)
- J-Link Software Package  
Provides drivers and tools for debugging and flash programming

※ Ozone should be installed first, followed by the J-Link Software and Documentation Pack.

## 2.4 Constraints (Debugging Limitations)

- GPIO10 and GPIO11 are dedicated to SWD. If these pins are used for other purposes (e.g., I2C, GPIO), J-Link debugging will not be available.
- J-Link debugging cannot be used in deep sleep scenarios involving microcode execution and firmware reload.

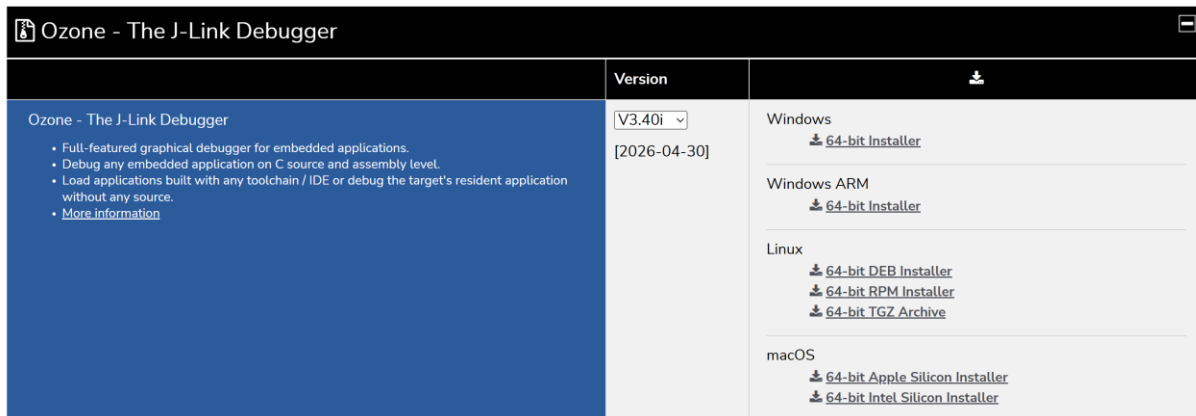
## 2.5 Reference Links

- J-Link Debugger:
  - <https://shop-us.segger.com/product/j-link-plus-classic-8-08-28/>
- Debug Probes:
  - <https://shop-us.segger.com/product-category/debug-probes/>
- Ozone IDE:
  - <https://www.segger.com/products/development-tools/ozone-j-link-debugger/>
- Ozone Manual:
  - [https://www.segger.com/downloads/jlink/UM08025\\_Ozone.pdf](https://www.segger.com/downloads/jlink/UM08025_Ozone.pdf) Installing Ozone IDE



### 3 Ozone installation

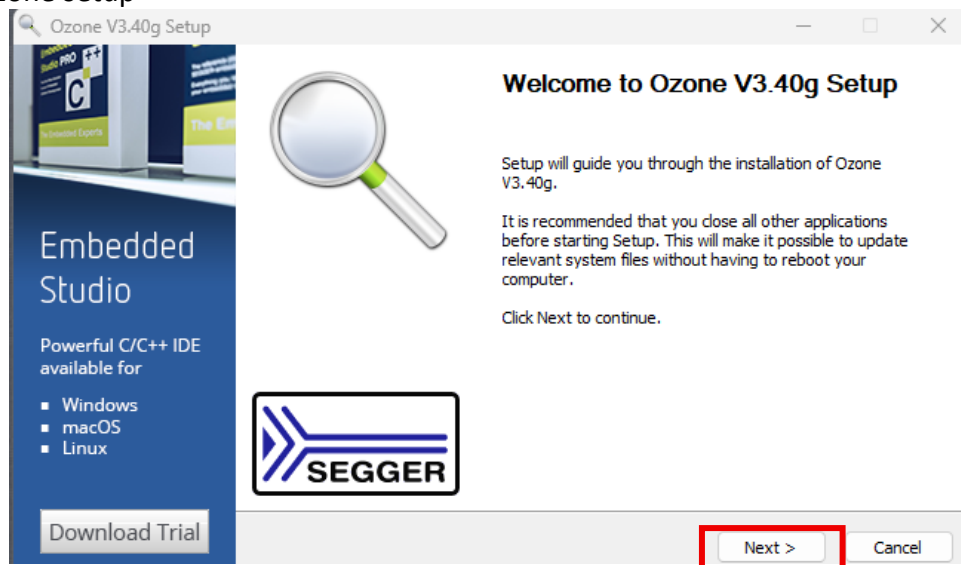
- Download Ozone from: <https://www.segger.com/downloads/jlink/#Ozone>



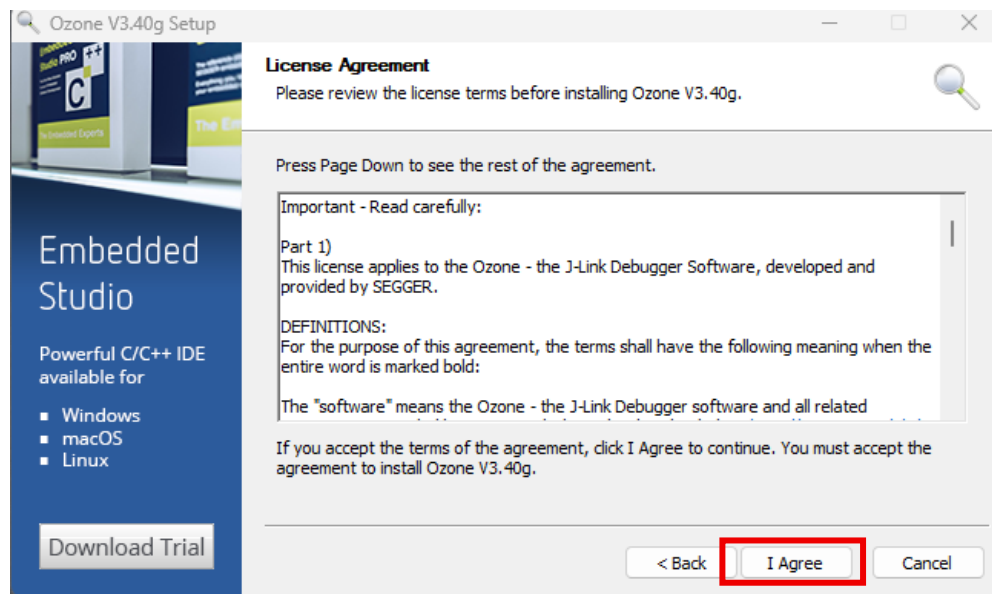
- Execute the Ozone install file

|                             |                   |             |           |
|-----------------------------|-------------------|-------------|-----------|
| Ozone_Windows_V340i_x64.exe | 5/27/2026 2:47 PM | Application | 17,861 KB |
|-----------------------------|-------------------|-------------|-----------|

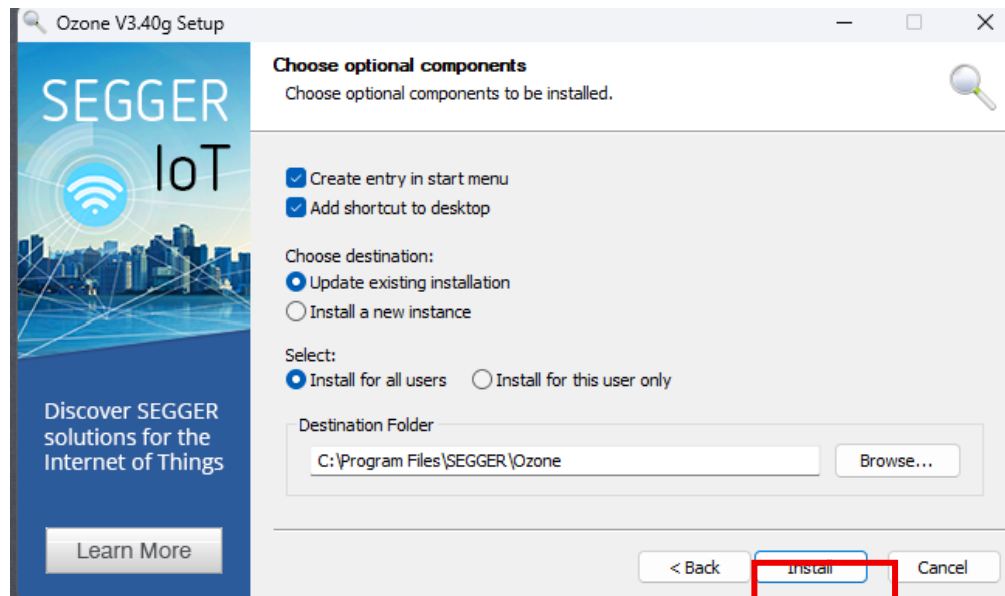
- Start Ozone setup



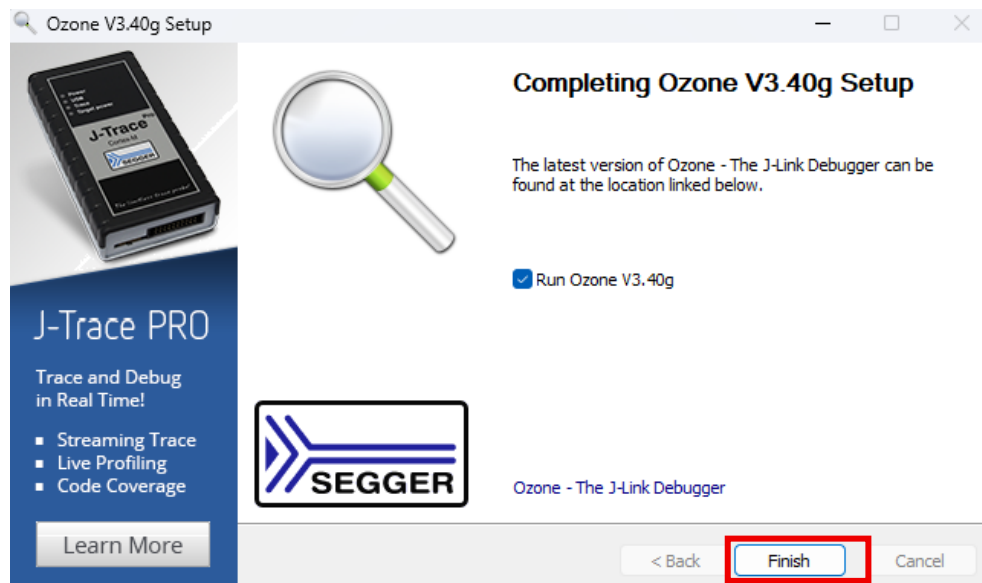
- License Agreement



- Choose optional components

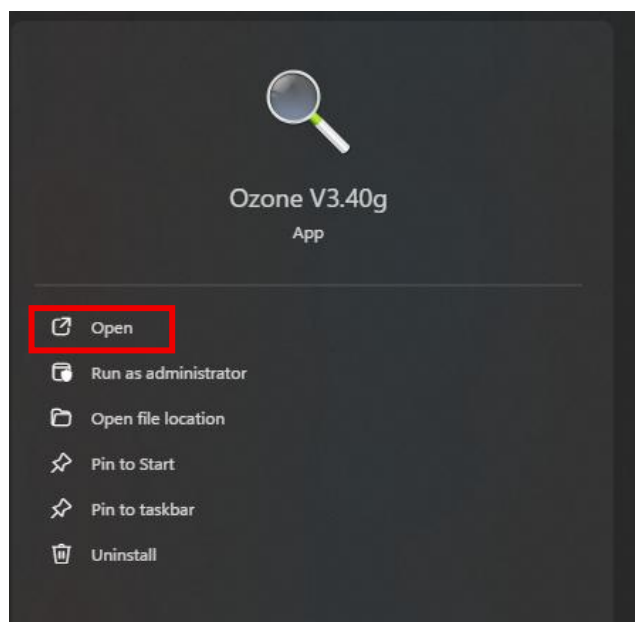


- Completing Ozone setup

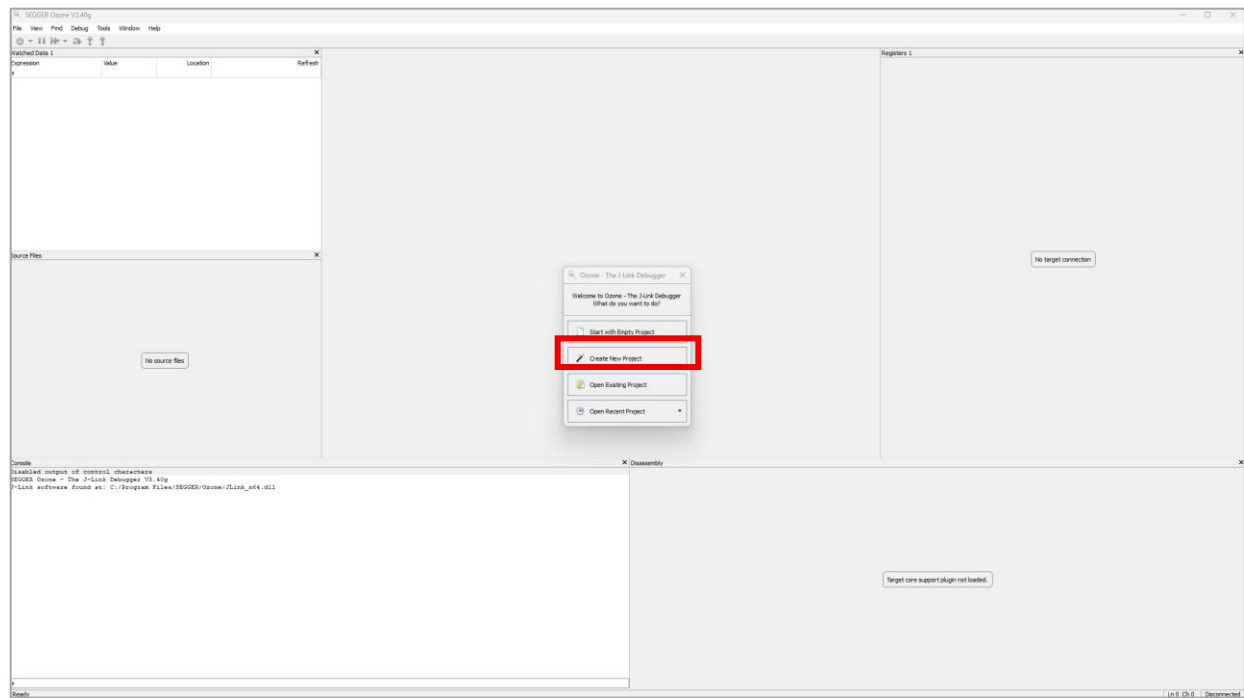


### 3.1 Creating a New Ozone Project

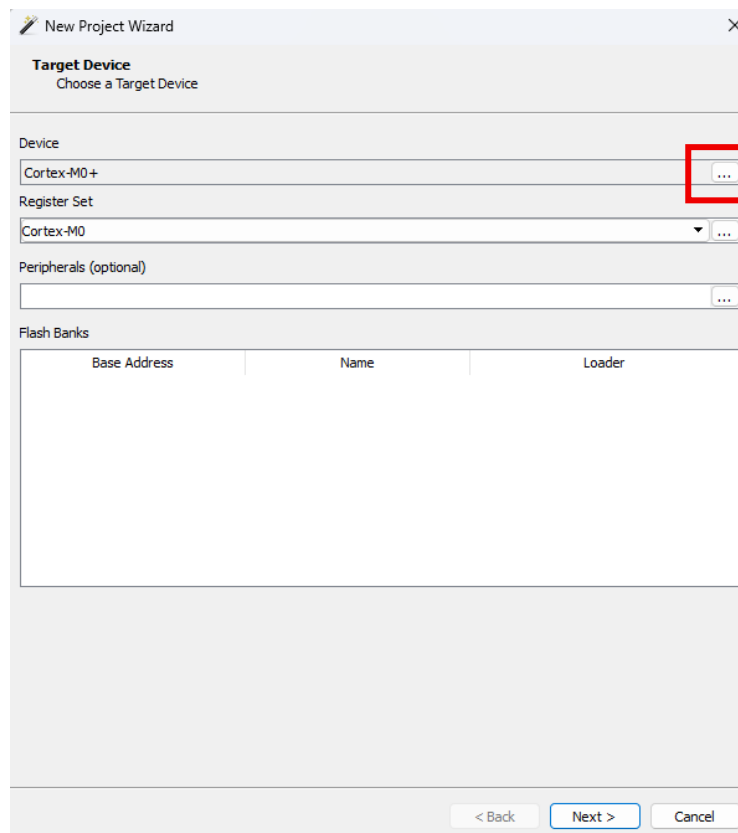
- Run the Ozone application after installation.
  - Before launching the Ozone application, ensure that the J-Link debugger is properly connected to the target evaluation board (EVB) via the SWD interface.



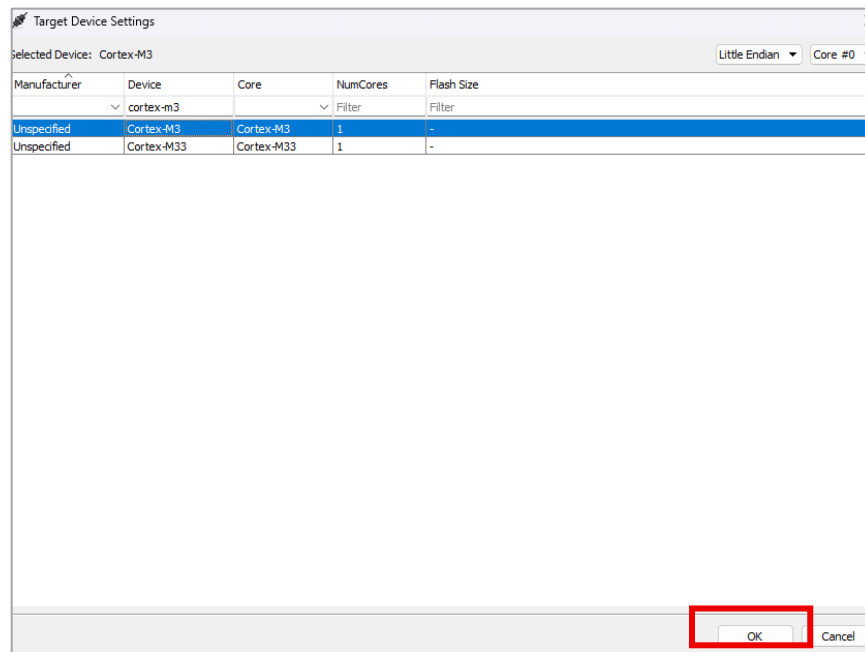
- Create New Project



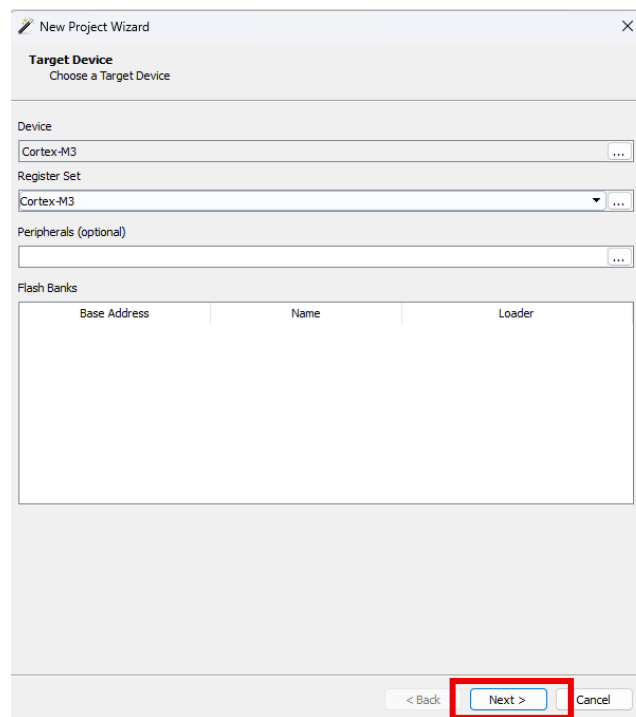
- Select target Device



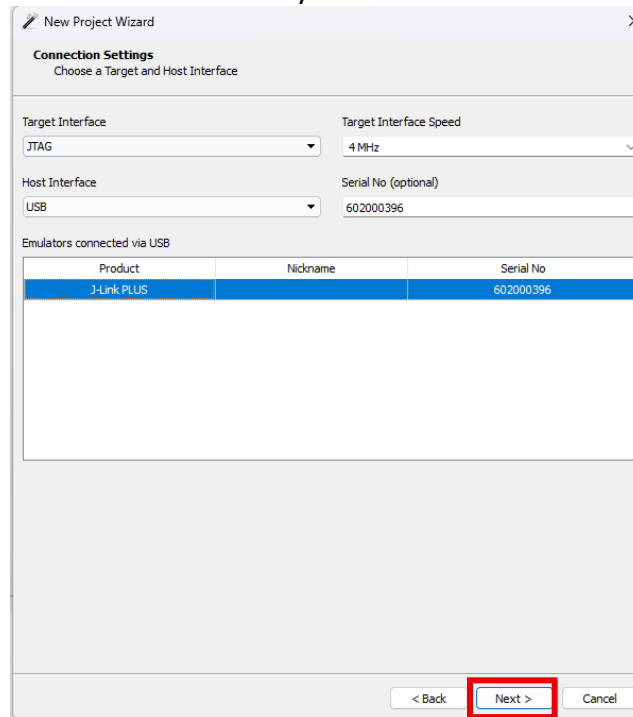
- Select target Device Settings to Cortex-M3



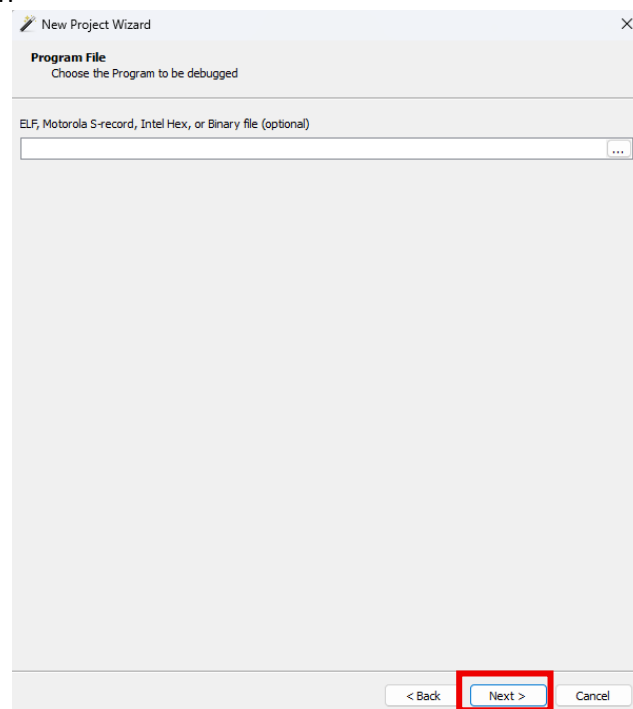
- Click the Next button



- Select the J-link and click the Next button
  - Select SWD as the target interface.
  - Select the product such as J-Link PLUS in the “Emulators connected via USB”. Serial No (optional) field is filled automatically.



- Click the Next button



- Click the Finish button

New Project Wizard

**Optional Settings**  
Set optional project settings, such as the initial PC

Initial PC (after download and reset)

☒ ELF Entry Point

☐ Read from Base Address Vector Table

☐ Read from Location

☐ Location

☐ Do not set

Initial Stack Pointer

☒ Read from Base Address Vector Table

☐ Read from Location

☐ Location

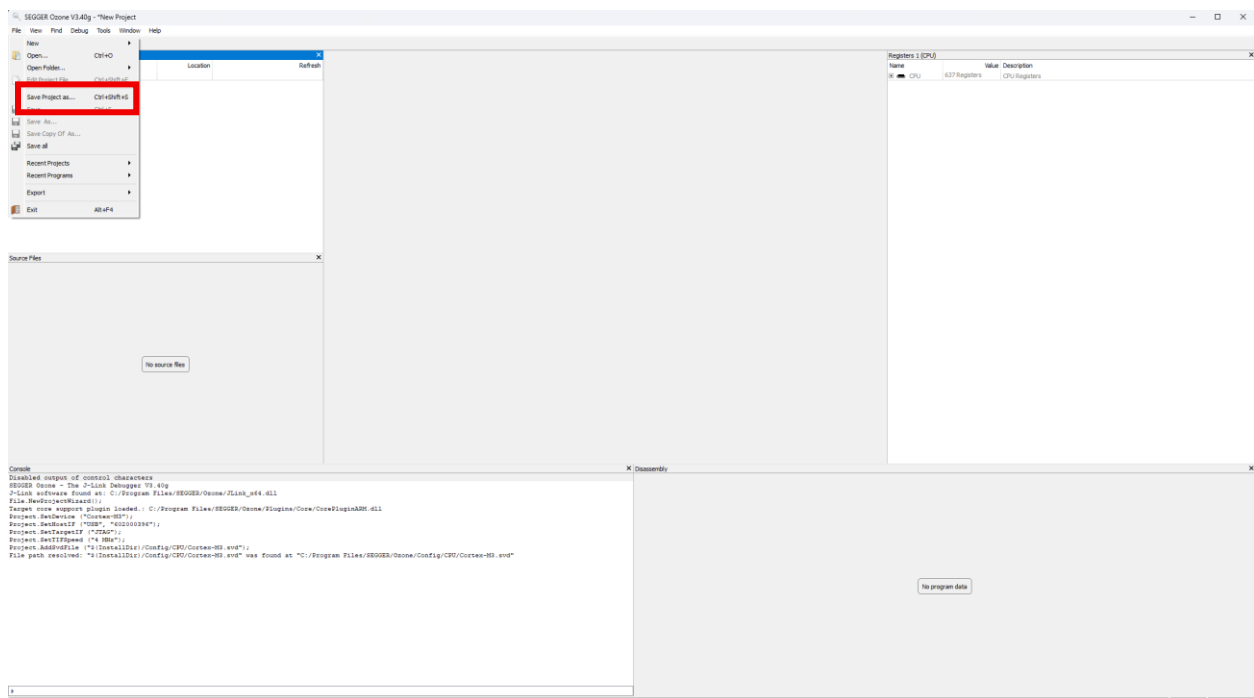
☐ Do not set

J-Link Script File

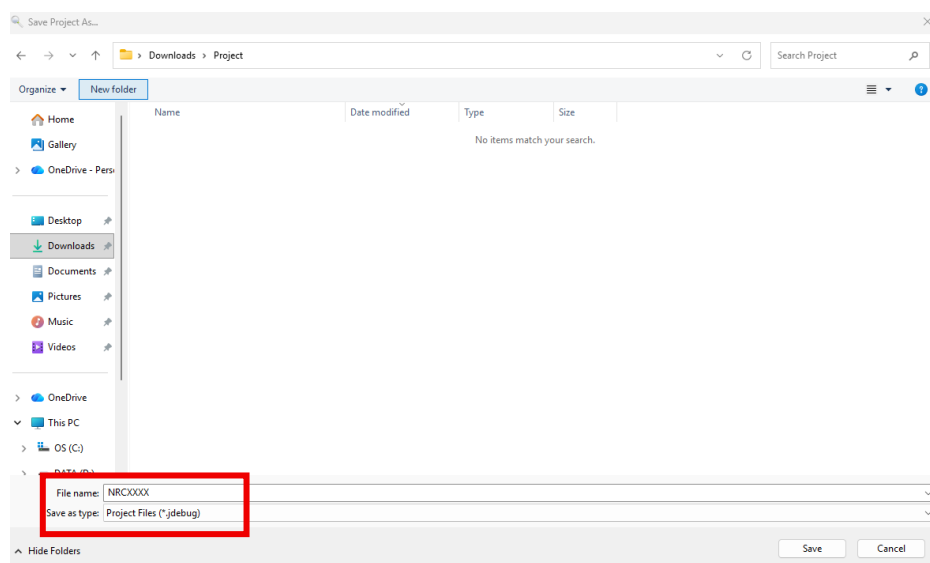
J-Link Log File

< Back Finish Cancel

- Sava Project



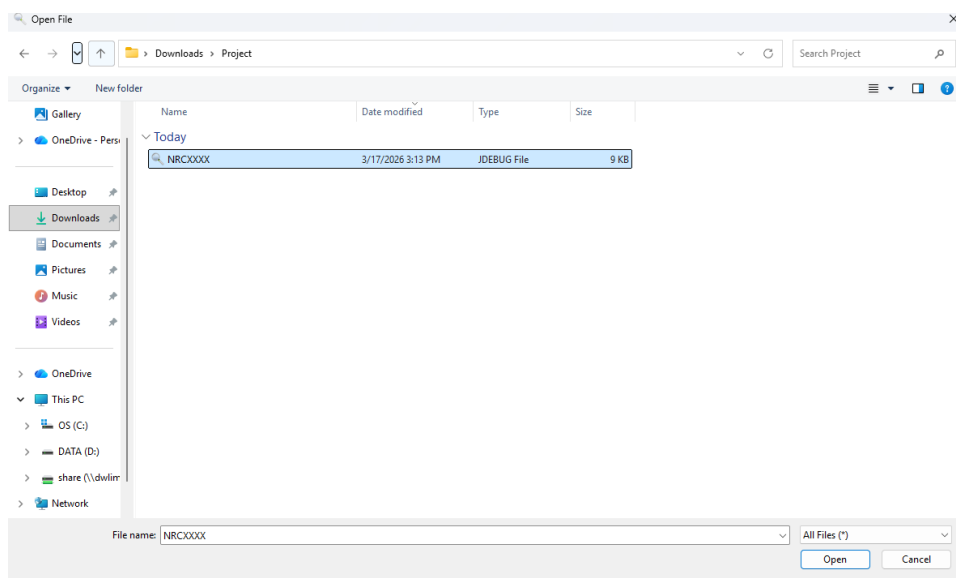
- Save Project with file name



Refer the 'project\_example.jdebug' file.

## 3.2 Project Configuration

- Open the generated .jdebug project file.
  - (File) – (Open)
- Select Project file





- Edit Project file

- (File) – (Edit Project File)
- Edit the **onProjectLoad()**

The following items must be updated in the Ozone project script according to the user's local development environment and target device configuration.

- **Project.SetDevice()**

This is updated to select the target name with NRCXXXX

- **Project.SetHostIF()**

The J-Link serial number must match the actually connected J-Link device.

For example, in the following configuration:

***Project.SetHostIF ("USB", "601018167");***

the value "601018167" is the J-Link serial number. If a different J-Link is connected, this value must be changed accordingly. The connected J-Link serial number can be checked using J-Link Commander or the SEGGER tools.

- **File.Open()**

This is updated to load the correct ELF file generated by the SDK build. The ELF file contains the application firmware image and debug symbols, so Ozone can map the source code, symbols, and breakpoints correctly during debugging.

```

/*****
*
*      OnProjectLoad
*
* Function description
*      Project load routine. Required.
*
*****/
void OnProjectLoad (void) {
    Project.SetDevice ("NRCXXXX");
    Project.SetHostIF ("USB", "601018167");
    Project.SetTargetIF ("SWD");
    Project.SetTIFSpeed ("4 MHz");
    Project.AddSvdFile ("$(InstallDir)/Config/CPU/Cortex-M3.svd");
    //
    // User settings
    //
    File.Open("\\wsl.localhost\Ubuntu-
22.04\home\pi\work\working\NRC_MACSW\out\nrc7394\standalone_xip\hello_world\nrc7394_standalone_xip_hello_w
orld_4M.elf");
}

```

- Only the application firmware will be debugged; `boot\_xip` is excluded. The target is allowed to run normally, and debugging is performed by attaching the debugger after execution has started.
- To ensure this behavior, the debugger script must be modified so that boot initialization is not executed. Specifically, comment out the `\_SetupTarget();` calls in both `AfterTargetDownload()` and `AfterTargetReset()`. This prevents `\_SetupTarget()` from being executed when those functions are called.

```
/******  
*  
*      AfterTargetReset  
*  
* Function description  
*   Event handler routine. Optional.  
*   The default implementation initializes SP and PC to reset values.  
**  
*****  
*/  
void AfterTargetReset (void) {  
    //  _SetupTarget();  
}  
  
/******  
*  
*      AfterTargetDownload  
*  
* Function description  
*   Event handler routine. Optional.  
*   The default implementation initializes SP and PC to reset values.  
*  
*****  
*/  
void AfterTargetDownload (void) {  
    //  _SetupTarget();  
}
```

## 4 J-LINK Installation

The J-Link Software and Documentation Pack, available in the below links

<https://www.segger.com/downloads/jlink/>

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Version               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>J-Link Software and Documentation Pack</b><br>Contains: <ul style="list-style-type: none"> <li>• Software</li> <li>• Manuals</li> <li>• Firmware updates</li> </ul> Can be downloaded and used free of charge by any owner of a SEGGER J-Link or J-Trace model. <ul style="list-style-type: none"> <li>• Not all features of it may be available on all J-Link / J-Trace models. Refer to the <a href="#">model overview</a> for more information.</li> <li>• Updated frequently</li> <li>• <a href="#">Release notes</a></li> <li>• <a href="#">More information</a></li> </ul> <input checked="" type="checkbox"/> Subscribe to J-Link software update notifications | V9.36<br>[2026-04-15] | <b>Windows</b><br><a href="#">64-bit Installer</a> <a href="#">32-bit Installer</a><br><br><b>Windows ARM</b><br><a href="#">64-bit Installer</a><br><br><b>Linux</b><br><a href="#">64-bit DEB Installer</a> <a href="#">32-bit DEB Installer</a><br><a href="#">64-bit RPM Installer</a> <a href="#">32-bit RPM Installer</a><br><a href="#">64-bit TGZ Archive</a> <a href="#">32-bit TGZ Archive</a><br><br><b>Linux ARM</b><br><a href="#">64-bit DEB Installer</a> <a href="#">32-bit DEB Installer</a><br><a href="#">64-bit TGZ Archive</a> <a href="#">32-bit TGZ Archive</a><br><br><b>macOS</b><br><a href="#">64-bit Apple Silicon Installer</a><br><a href="#">64-bit Intel Silicon Installer</a><br><a href="#">64-bit Universal Installer</a> |

It provides the necessary tools for flash programming, including J-Flash and J-Link Commander. Using these tools, firmware images can be written to the target device's flash memory.

After installation, copy FlashNRCXXXX.elf and JLinkDevices.xml into the J-Link installation directory.

(ex) C:\Program Files\SEGGER\JLink\_V938a

- FlashNRCXXXX.elf  
This file is a flash loader (flash algorithm) used by J-Link to access and program the external SPI flash. It contains the routines required for flash initialization, erase, write, and verify operations.
- JLinkDevices.xml  
This file defines the target device configuration for J-Link. It registers the NRC device, including memory layout and flash information, and links the flash loader (FlashNRCXXXX.elf) to the device so that J-Link can correctly recognize and program the target.

## 5 Prepare the Firmware Configuration

To enable stable J-Link debugging, certain runtime protections in the firmware must be disabled.

By default, the system performs:

- Firmware size and CRC validation using FW\_INFO
- Watchdog reset during firmware execution

These mechanisms can interfere with debugging (e.g., unexpected reset, halted execution), so they must be disabled before attaching the debugger.

### 5.1.1 Update the boot\_xip binary

Before using the boot configuration feature for J-Link debugging, the `boot\_xip` binary must be updated at least once.

The updated XIP bootloader includes support for:

- Boot configuration (`boot\_config`)
- Skipping firmware size and CRC validation during J-Link debugging

Without this updated bootloader, the system will continue performing FW\_INFO-based firmware integrity verification during XIP boot.

The updated functionality is included in the following FirmwareFlashTool versions or later:

v6.5.1 / v7.4.2 (05/07/2026)

- Updated bootloader/firmware
- Added boot\_config support for J-Link in XIP bootloader

When updating the boot\_xip binary using FirmwareFlashTool:

- Enable the Upload checkbox in the XIP BOOTLOADER PATH (.bin) section
- Select the updated nrc7394\_boot\_xip.bin image
- Click START to program the updated bootloader

Please update the boot\_xip binary before using J-Link debugging with boot\_config.

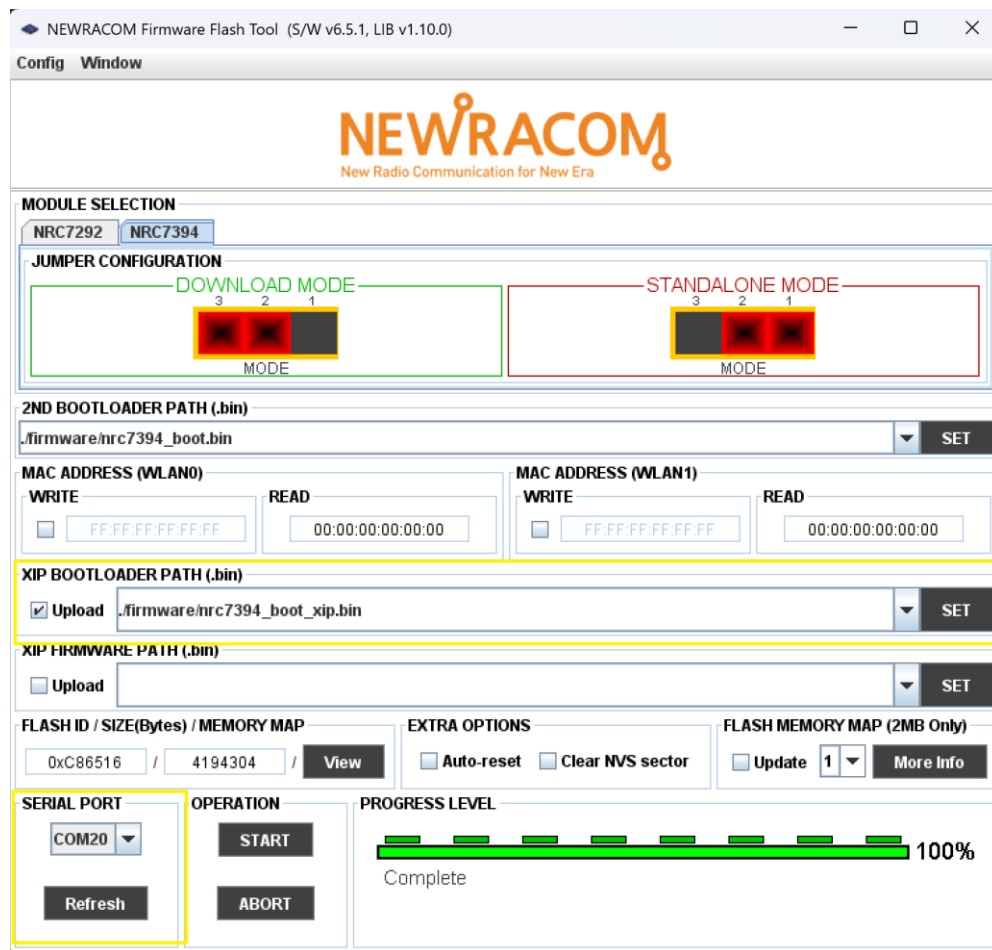


Figure 5.1 Firmware Uploader Tool

### 5.1.2 boot\_config.json Description

The boot\_config.json file is used to configure boot-time behavior of the system. It allows overriding specific runtime checks by writing key-value pairs into flash.

Example configuration:

- jlink\_enable : Enables J-Link debugging.
- wdt\_disable : Disables the watchdog timer. Prevents unexpected system resets during debugging
- crc\_check\_skip : Skips FW\_INFO-based firmware size and CRC validation. Allows execution without firmware integrity checks

```
{
  "jlink_enable": 1,
  "wdt_disable": 1,
  "crc_check_skip": 1
}
```

### 5.1.3 Batch Script Execution Guide

- Usage:
  - Reset the board in XIP write (download) mode and disconnect the console terminal to avoid communication conflicts.
  - Type the following command in Command Prompt or PowerShell:

```
nrc7394_boot_config.exe <serial port> <boot configuration JSON path>
```

```
D:\nrc7394_boot_config_v1_0>nrc7394_boot_config.exe com15 boot_config.json
[PROGRESS] Uploading the 2nd bootloader.
[PROGRESS] Running the 2nd bootloader.
[PROGRESS] Updating the boot configuration.
```

- Switch to XIP mode, reconnect the console terminal, and perform a hardware reset on the EVB.
  - If the configuration is applied successfully, you can verify it by checking the console logs, where messages related to fw\_config.json will be printed.

```
CRC check skipped by boot config.
[boot_cfg] jlink_enable=1 wdt_disable=1 crc_check_skip=1
Cold Boot!! Clear all info in retent Info <Total: 15875 Byte Cleared>
[POR]
[system_modem_api_read_fc_common_slot] fail to read sta data <slot>
Initial CAL is done
[RF CAL] loaded 1032 bytes <data_count=2> from FLASH
[RF CAL] selected data index 0 <cache>
[set_txpwr] lpf=0xB, lo_param0=-15, lo_param1=-6
Init MCS(-1) for Null data
[!] evt_mgr: : nrc_event_mgr_init done: queue=0x10438538 task=0x10440e48
[!mac_ps_set_dns_server] ipaddr1: 08080808, ipaddr2: 0404win: 0 MAC A0808
```

#### ※ Important Note

If the XIP firmware is updated using FirmwareFlashTool, the boot\_config area stored in flash will be erased. In this case, the boot\_config data must be written again using nrc7394\_boot\_config.exe.



- **L1-Verify (Level 1 Verify):** A fast verification step that checks the programmed data, typically using a quick method like read-back comparison or checksum to ensure correctness.
  - **L2-Verify**  
This is a more thorough and complete verification step. It performs a full read-back and byte-by-byte comparison of the flash contents against the original data to guarantee data integrity.
- ※ After downloading the firmware, perform a hardware reset to start normal execution. Then, proceed with debugging using **Attach & Halt Program** to attach to the running system.

### 6.1.2 Attach & Halt Program

This method attaches the debugger to a target that is already running and halts execution at the current state.

- Let the firmware run normally
- Click **Attach & Halt Program**
- The debugger connects and halts the CPU

When the debugger is connected using **Attach & Halt Program**, the CPU is automatically halted at the moment of connection.

To resume execution for debugging:

- Select **(Debug) → (Continue)** to allow the firmware to run.



## 7 Revision history

| Revision No | Date      | Comments        |
|-------------|-----------|-----------------|
| Ver 1.0     | 5/28/2026 | Initial version |
|             |           |                 |
|             |           |                 |
|             |           |                 |